

NRP Case



Section I: Scenario Demographics

Scenario Title:	NRP
Date of Development:	29/06/2015 (DD/MM/YYYY)
Target Learning Group:	☐ Juniors (PGY 1 - 2) ☐ Seniors (PGY $\geq$ 3) ☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
Affiliations/Institution(s):	McMaster University
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	Expose learners to the resuscitation of a neonate – a rare but critical presentation in the ED.
CRM Objectives:	1) Effectively lead a team through a neonatal resuscitation. 2) Demonstrate high quality closed-loop communication during resuscitation.
Medical Objectives:	1) Recognize flat baby and initiate warming and drying. 2) Provide quality NRP care including: a. Starting with PPV b. Initiating CPR when appropriate c. Intubation d. Attempting iv access/umbilical line

Case Summary: Brief Summary of Case Progression and Major Events
The team has been called to help in the ED where a woman just precipitously gave birth to a baby now requiring resuscitation. The mom thinks she's at term. She has had no prenatal care and is an iv drug user. The baby will be flat. After stimulation and drying, the baby will have a HR <100 and PPV will be required. After 60 seconds, the HR will still be <60 and CPR will need to be started. This will be short lived. The team will also need to intubate and obtain IV access.

Facilitators Required to Run Session
Instructors (faculty or senior resident): 1 (to observe). Could add second to play role of other ED MD (to provide more history re: mother and pregnancy)
Confederate nurse: 1 (to assist at bedside, describe neonate as flat)
Sim tech: 1



Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates		Brief Description of Role	
Nurse		To assist at bedside, describe infant's status as flat, and provide history.	
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Neonatal Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☒ Other: Radiant warmer	
☐ Blood Products	☒ Neonatal ET Tubes	☒ Other: Umbilical Vein Catheter set-up	
☒ Intraosseous Set-up	☐ LMA	☒ Other: Blankets for drying	
D. Moulage			
None required.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	10 min
		Debriefing:	20 min



Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

You are working in the minor area of your ED and have been called by the physician on the major side to assist with a precipitous delivery. He is managing the mother and wants you to be ready to resuscitate the infant if needed. The mom thinks she's term. She's had no prenatal care and is an IV drug user. She used earlier today. There no meconium staining noted in the amniotic fluid. Baby has just been delivered and is handed to your team.

B. Patient Profile and History

Patient Name: Baby Girl Adams	Age: 0 days	Weight: 2.5kg (5 lbs 8 oz)
Gender: ☐ M ☒ F		
Chief Complaint: Flat baby		
History of Presenting Illness: Precipitous delivery to an IVDU		
Past Medical History:	NA	Medications: NA
Allergies: NA		
Social History: Born to IVDU. Child protective services not yet involved. (No prenatal care for mom.)		
Review of Systems:	CNS:	NA
	HEENT:	NA
	CVS:	NA
	RESP:	NA
	GI:	NA
	GU:	NA
	MSK:	NA

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 50/min	BP: ?	RR: 20/min O ₂ SAT: 58%
Rhythm: Sinus brady	T: 36.5°C	Glucose: 5.2 mmol/L
General Status: Flat. Not crying. Poor tone and respiratory effort.		
CNS:	Poor tone. Flat. Not crying.	
HEENT:	Nil.	
CVS:	HR 80.	
RESP:	Poor respiratory effort.	
ABDO:	Soft, NT.	
GU:	Nil.	
MSK:	Nil.	SKIN: Gray-looking. Not pink.



Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus HR: 50/min BP: ?/? RR: 20/min O ₂ SAT: 58% T: 36.5°C	Flat. Not crying. Minimal respiratory effort.	<u>Learner Actions</u> - ☐ Stimulate, warm & dry - ☐ Bring to neonatal warmer - ☐ Check HR → start PPV with 21% FiO ₂ - ☐ Consider suctioning - ☐ Initiate pre-ductal SpO ₂ monitoring (right arm) - ☐ Check cap sugar: 5.2	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - If naloxone given → seizure <u>Triggers</u> <i>For progression to next state</i> - 1 min → 2. 60 Second Mark
2. 60 Second Mark Unchanged	Unchanged.	<u>Learner Actions</u> - ☐ Re-check HR at 60 sec - ☐ Optimize ventilation, switch to 100% FiO ₂ - ☐ Intubate - ☐ Start CPR at 90 sec or after intubation - ☐ Maintain 3:1 ratio for CPR using thumb encircling or two finger technique - ☐ Check HR/RR q1min	<u>Modifiers</u> - 2 min → O ₂ SAT 65% - 3 min → O ₂ SAT 70% <u>Triggers</u> - 4 min → 3. HR >60
3. HR > 60 HR → 80 RR → 40 O ₂ SAT → 75% BP → 55/30	Appears less grey.	<u>Learner Actions</u> - ☐ Obtain iv access - ☐ Consider IVF bolus (10ml/kg =25ml) - ☐ Check cap sugar (if not yet) - ☐ Send blood work, including blood cultures - ☐ Prepare for UVC placement	<u>Modifiers</u> - 5 min → O ₂ SAT 80% <u>Triggers</u> - 7 min → 4. HR >100 (set to slowly return to normal)
4. HR >100 O ₂ SAT → 83% HR → 120	Appears pink. Moving limbs now.	<u>Learner Actions</u> - ☐ Call NICU - ☐ Insert UVC - ☐ AXR to confirm UVC placement, CXR for ETT - ☐ Finish tasks from above	END CASE at 10 minutes or once intubated/UVC placed.



Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results
None for this case. No imaging for case.

Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) Have you ever taken part in NRP before? How did it feel today? 2) What are the key moments in the NRP algorithm? 3) How would your management have changed if the infant had meconium? 4) When is naloxone indicated in a neonate? What happens when you give naloxone to a neonate whose mom has been using narcotics regularly throughout pregnancy? 5) Were you concerned about the O ₂ sats? What is normal? Where should you measure them? 6) How do you feel the team communicated? Were you all being clear with each other? How did you address uncertainty in management?	
Key Moments	
Recognition of flat baby.	
Initiating PPV.	
Initiating CPR.	